

AYDIN ADNAN MENDERES UNIVERSITY

CSE 431

NATURAL LANGUAGE PROCESSING WITH MACHINE LEARNING TERM PROJECT

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Objective

This project develops a Turkish smart home voice command recognition system using Word2Vec embeddings. The system interprets user speech, understands semantic variations, and maps them to predefined commands. The goal is to compare different sentence embedding techniques to determine the most effective approach for command recognition.

Problem Statement:

Natural language understanding faces challenges when dealing with:

- **Semantic variations:** Users express the same intent differently ("ışığı yak" vs "lambayı aç")
- **Unknown words:** Slang or misspellings ("şavkı" for "ışığı")
- **Out-of-vocabulary terms:** Words not present in pre-trained models

Traditional word-independent approaches fail to capture phrase-level semantics, leading to false matches based on common verbs alone.

Data Set Creation and Model Training

```
Reading core commands from Excel...
Dataset created successfully!
Core commands: 71
Commands with variations: 71
Total variations generated: 193
Total sentences: 264
Saved to: TR_Commands_Expanded.xlsx
Text Preprocessing
Preprocessed 264 sentences
Building N-gram Model
N-gram transformation examples:
'ışığı aç' → 'ışığı_aç'
'lambayı aç' → 'lambayı_aç'
'ışıkları aç' → 'ışıkları_aç'
'aydınlatmayı çalıştır' → 'aydınlatmayı_çalıştır'
'ışıkları yak' → 'ışıkları_yak'
Total bigrams detected: 245
Loading Pre-trained Turkish Word2Vec Model
Found existing 'tvec.txt'
Training Word2Vec Models
Pre-trained vector dimension: 400

Model 1: Word-level embeddings (without n-grams)
Vocabulary size: 172 words
Transferred pre-trained vectors: 157 (91.3%)
Training for 20 epochs...
Saved: smart_home_model.model
```

```
Model 2: N-gram embeddings (with bigrams)
Vocabulary size: 254 tokens
Transferred pre-trained vectors: 32 (12.6%)
Training for 20 epochs...
Saved: smart_home_model_ngram.model

Generated files:
TR_Commands_Expanded.xlsx (264 sentences)
smart_home_model.model (word-level)
smart_home_model_ngram.model (n-gram)
```

- For each core command, semantically similar phrases were generated
- Training Dataset:
 - Core commands: 71
 - Total variations: 193
 - Total Training Sentences: 264
- Pre-trained model: Turkish Word2Vec
- Fine-tuning: 20 epochs on smart home corpus
- 2 models trained:
 1. Word-level model: Individual word embeddings
 2. N-gram model: Phrase-level embeddings
- Sentence Embedding Methods:
 1. Mean of Word Vectors
 2. TF-IDF Weighted Mean

Testing with Word Level Model

```

Your Command: ışıklarını aç
Method 1 (Mean):  ışığı aç (Score: 1.00)
Method 2 (TF-IDF):  ışığı aç (Score: 1.00)

Your Command: lambayı yak
Method 1 (Mean):  ışığı yak (Score: 1.00)
Method 2 (TF-IDF):  ışığı yak (Score: 1.00)

Your Command: televizyonu aç
Method 1 (Mean):  tv aç (Score: 1.00)
Method 2 (TF-IDF):  tv aç (Score: 1.00)

Your Command: klimayı çalıştır
Method 1 (Mean):  klimayı aç (Score: 1.00)
Method 2 (TF-IDF):  klimayı aç (Score: 1.00)

Your Command: aç
Method 1 (Mean):  multimedyaı aç (Score: 1.00)
Method 2 (TF-IDF):  televizyonu aç (Score: 1.00)

Your Command: kapat
Method 1 (Mean):  tv kapa (Score: 1.00)
Method 2 (TF-IDF):  tv kapa (Score: 1.00)

Your Command: yak
Method 1 (Mean):  ışığı yak (Score: 1.00)
Method 2 (TF-IDF):  ışığı yak (Score: 1.00)

```

- Single verb "aç" matches different commands with perfect confidence depending on the method. System cannot determine which device the user wants to open.
- Isolated verbs match with 1.00 confidence. The system cannot distinguish between "tv'yi kapat", "ışığı kapat", or "klimayı kapat" when user only says "kapat".
- The word-level model treats words independently. When input contains only a verb, that single word's vector represents the entire command.

Testing with N-grams Model

```
Your Command: ışıklarını aç
Transformed to: ['ışıklarını_aç']
Method 1 (Mean):    ışığı aç (Score: 1.00)
Method 2 (TF-IDF):  ışığı aç (Score: 1.00)

Your Command: lambayı yak
Transformed to: ['lambayı_yak']
Method 1 (Mean):    ışığı yak (Score: 1.00)
Method 2 (TF-IDF):  ışığı yak (Score: 1.00)

Your Command: televizyonu aç
Transformed to: ['televizyonu_aç']
Method 1 (Mean):    tv aç (Score: 1.00)
Method 2 (TF-IDF):  tv aç (Score: 1.00)

Your Command: klimayı çalıştır
Transformed to: ['klimayı_çalıştır']
Method 1 (Mean):    klimayı aç (Score: 1.00)
Method 2 (TF-IDF):  klimayı aç (Score: 1.00)

Your Command: aç
Transformed to: ['aç']
Method 1 (Mean):    multimedyaı aç (Score: 1.00)
Method 2 (TF-IDF):  multimedyaı aç (Score: 1.00)
```

```
Your Command: kapat
Transformed to: ['kapat']
Method 1 (Mean):    multimedyaı kapa (Score: 1.00)
Method 2 (TF-IDF):  multimedyaı kapa (Score: 1.00)

Your Command: yak
Transformed to: ['yak']
Method 1 (Mean):    NOT RECOGNIZED (Score: 0.00)
Method 2 (TF-IDF):  NOT RECOGNIZED (Score: 0.00)

Your Command: söndür
Transformed to: ['söndür']
Method 1 (Mean):    NOT RECOGNIZED (Score: 0.00)
Method 2 (TF-IDF):  NOT RECOGNIZED (Score: 0.00)

Your Command: buzdolabını aç
Transformed to: ['buzdolabını', 'aç']
Method 1 (Mean):    multimedyaı aç (Score: 1.00)
Method 2 (TF-IDF):  multimedyaı aç (Score: 1.00)

Your Command: çamaşır makinasını çalıştır
Transformed to: ['çamaşır', 'makinasını', 'çalıştır']
Method 1 (Mean):    NOT RECOGNIZED (Score: 0.00)
Method 2 (TF-IDF):  NOT RECOGNIZED (Score: 0.00)

Your Command: kapıyı kilitle
Transformed to: ['kapıyı', 'kilitle']
Method 1 (Mean):    NOT RECOGNIZED (Score: 0.00)
Method 2 (TF-IDF):  NOT RECOGNIZED (Score: 0.00)
```

- The n-gram model still accepts isolated verbs with high confidence, similar to the word-level model. This occurs because single words like "aç" and "kapat" exist in the vocabulary as standalone tokens.
- "yak" and "söndür" isolated verbs are properly rejected. The difference in behavior suggests "yak" and "söndür" may not appear as standalone tokens in the training corpus, while "aç" and "kapat" do.
- Out-of-domain device "buzdolabını" is not in vocabulary. The model ignores the unknown word and matches based on "aç" alone, resulting in a false positive.
- Input: "çamaşır makinasını çalıştır"
- Transformed to: ['çamaşır', 'makinasını', 'çalıştır']
- Result: NOT RECOGNIZED (Score: 0.00)
- Input: "kapıyı kilitle"
- Transformed to: ['kapıyı', 'kilitle']
- Result: NOT RECOGNIZED (Score: 0.00)
- Out-of-domain phrases properly rejected.
- The n-gram model provides partial improvement over word-level embeddings. It successfully treats complete phrases as atomic units and rejects some isolated verbs ("yak", "söndür"). However, it still suffers from false positives when encountering common standalone verbs ("aç", "kapat") or when unknown words are paired with known verbs. The model's effectiveness depends heavily on how the training corpus structures individual words versus phrase units.

Evaluation Models

Test Dataset:

- 19 correct commands (variations of training data)
- 20 incorrect commands (out-of-domain phrases)
- Total: 39 test cases

Metrics:

- Precision: Of accepted commands, how many were correct?
- Recall: Of valid commands, how many did we catch?
- F1-Score: Harmonic mean balancing both metrics

METHOD 1 (Mean of Vectors - No N-grams) :

```
EVALUATING: Method 1: Mean of Vectors (No N-grams)

MISMATCHES (9 errors):
'volümü artır'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'sesi alçalt'
  REJECTED (score: 0.646) | SHOULD ACCEPT | Matched: tv aç
'sessize yaklaş'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'müzik çal'
  ACCEPTED (score: 0.932) | SHOULD REJECT | Matched: müzik kapa
'pencereyi aç '
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: perdeyi aç
'duş yap'
  ACCEPTED (score: 0.977) | SHOULD REJECT | Matched: ışığı kıs
'kahve yap'
  ACCEPTED (score: 0.977) | SHOULD REJECT | Matched: ışığı kıs
'alışveriş yap'
  ACCEPTED (score: 0.977) | SHOULD REJECT | Matched: ışığı kıs
'film izle'
  ACCEPTED (score: 0.864) | SHOULD REJECT | Matched: film zamanı

METRICS:
Precision: 0.727 (of accepted commands, 72.7% were correct)
Recall:    0.842 (caught 84.2% of valid commands)
F1-Score: 0.780
```

CONFUSION MATRIX:

	Predicted NO	Predicted YES
Actual NO (TN/FP)	14	6
Actual YES (FN/TP)	3	16
True Negatives (TN):	14 - Correctly rejected invalid commands	
False Positives (FP):	6 - Incorrectly accepted invalid commands	
False Negatives (FN):	3 - Incorrectly rejected valid commands	
True Positives (TP):	16 - Correctly accepted valid commands	

- High recall (84.2%) - catches most valid commands despite vocabulary gaps
- Accepts semantically similar but incorrect commands

METHOD 2 (Mean of Vectors With N-grams) :

```
MISMATCHES (11 errors):
'savkı yak'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'ekranı karart'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'volümü artır'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'sesi alçalt'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'odayı aydınlat'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'aydınlanmayı durdur'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'sessize yaklaş'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'pencereyi aç '
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: perdeyi aç
'duş yap'
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: ışığı arttır
'kahve yap'
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: ışığı arttır
'alışveriş yap'
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: ışığı arttır

METRICS:
Precision: 0.750 (of accepted commands, 75.0% were correct)
Recall:    0.632 (caught 63.2% of valid commands)
F1-Score:  0.686
```

CONFUSION MATRIX:

	Predicted NO	Predicted YES
Actual NO (TN/FP)	16	4
Actual YES (FN/TP)	7	12
True Negatives (TN):	16 - Correctly rejected invalid commands	
False Positives (FP):	4 - Incorrectly accepted invalid commands	
False Negatives (FN):	7 - Incorrectly rejected valid commands	
True Positives (TP):	12 - Correctly accepted valid commands	

- Higher precision (75.0%) - more conservative, fewer false positives
- Lower recall (63.2%) - rejects valid variations not in training bigrams

METHOD 3 (TF-IDF Weighted - No N-grams) :

```
EVALUATING: Method 3: TF-IDF Weighted (No N-grams)

MISMATCHES (9 errors):
'volümü artır'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'sesi alçalt'
  REJECTED (score: 0.641) | SHOULD ACCEPT | Matched: tv aç
'sessize yaklaş'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'müzik çal'
  ACCEPTED (score: 0.922) | SHOULD REJECT | Matched: müzik kapa
'pencereyi aç '
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: perdeyi aç
'duş yap'
  ACCEPTED (score: 0.948) | SHOULD REJECT | Matched: ışığı kıs
'kahve yap'
  ACCEPTED (score: 0.948) | SHOULD REJECT | Matched: ışığı kıs
'alışveriş yap'
  ACCEPTED (score: 0.948) | SHOULD REJECT | Matched: ışığı kıs
'film izle'
  ACCEPTED (score: 0.921) | SHOULD REJECT | Matched: film zamanı

METRICS:
Precision: 0.727 (of accepted commands, 72.7% were correct)
Recall:    0.842 (caught 84.2% of valid commands)
F1-Score:  0.780
```

CONFUSION MATRIX:

	Predicted NO	Predicted YES
Actual NO (TN/FP)	14	6
Actual YES (FN/TP)	3	16
True Negatives (TN):	14 - Correctly rejected invalid commands	
False Positives (FP):	6 - Incorrectly accepted invalid commands	
False Negatives (FN):	3 - Incorrectly rejected valid commands	
True Positives (TP):	16 - Correctly accepted valid commands	

- TF-IDF weighting provides minimal benefit at word level. Common verbs ("aç", "kapat", "yap") get lower weights, distinctive words ("volümü", "pencereyi") get higher weights, but overall performance is unchanged.

METHOD 4 (TF-IDF Weighted With N-grams) :

```
MISMATCHES (11 errors):
'savkı yak'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'ekranı karart'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'volümü artır'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'sesi alçalt'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'odayı aydınlat'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'aydınlanmayı durdur'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'sessize yaklaş'
  REJECTED (score: 0.000) | SHOULD ACCEPT | Matched: ışığı aç
'pencereyi aç '
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: perdeyi aç
'duş yap'
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: ışığı arttır
'kahve yap'
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: ışığı arttır
'alışveriş yap'
  ACCEPTED (score: 1.000) | SHOULD REJECT | Matched: ışığı arttır

METRICS:
Precision: 0.750 (of accepted commands, 75.0% were correct)
Recall:    0.632 (caught 63.2% of valid commands)
F1-Score:  0.686
```

```
CONFUSION MATRIX:

                Predicted NO    Predicted YES
Actual NO  (TN/FP)           16             4
Actual YES (FN/TP)           7             12

True Negatives (TN):    16 - Correctly rejected invalid commands
False Positives (FP):   4  - Incorrectly accepted invalid commands
False Negatives (FN):   7  - Incorrectly rejected valid commands
True Positives (TP):    12 - Correctly accepted valid commands
```

- TF-IDF provides no additional benefit when using n-grams. Since bigrams are already atomic semantic units ("ışığı_aç", "lambayı_yak"), weighting individual n-grams doesn't improve discrimination.

COMPARATIVE ANALYSIS

FINAL COMPARISON TABLE

Method	Precision	Recall	F1-Score	Errors
Mean (No N-grams)	0.727	0.842	0.780	9
Mean (With N-grams)	0.750	0.632	0.686	11
TF-IDF (No N-grams)	0.727	0.842	0.780	9
TF-IDF (With N-grams)	0.750	0.632	0.686	11
BEST METHOD: Mean (No N-grams)				
F1-Score: 0.780				
Precision: 0.727				
Recall: 0.842				
Method	TP	TN	FP	FN
Mean (No N-grams)	16	14	6	3
Mean (With N-grams)	12	16	4	7
TF-IDF (No N-grams)	16	14	6	3
TF-IDF (With N-grams)	12	16	4	7

Best Method: Method 1 & 3 (Word-Level approaches) with F1: 0.780

- Word-level embeddings outperform n-gram embeddings in overall F1-score
- N-grams improve precision (+2.3%) but significantly reduce recall (-21.0%)
- TF-IDF weighting shows no measurable improvement over simple mean vectors
- Word-level accepts more commands (good recall) but less selective (lower precision)

For production systems, choose based on use case:

- High recall needed (minimize user frustration): Use Method 1 (Mean, Word-Level)
- High precision needed (minimize wrong activations): Use Method 2 (Mean, N-grams)

Technical Methodology

Word Embeddings (400 Dimensions)

- Word2Vec represents each word as a 400-dimensional vector in mathematical space. Each dimension captures different semantic features of the word. For example, dimensions might encode properties like "is it an action?", "related to light?", "related to sound?". The 400-dimensional space allows the model to capture rich semantic relationships between words. Words with similar meanings (like "ışık" and "lamba") will have vectors pointing in similar directions in this high-dimensional space.

Sentence Embeddings

- To represent entire commands, we aggregate word vectors using two approaches:
 - Mean Vector: Arithmetic average of all word vectors in the command
 - TF-IDF Weighted: Weighted average where important words (high TF-IDF) contribute more to the final vector

Cosine Similarity

- Cosine similarity measures how similar two vectors are by calculating the cosine of the angle between them. The result ranges from -1 to 1:
 - 1.0 = identical meaning (vectors point in same direction)
 - 0.0 = unrelated (vectors are perpendicular)
 - -1.0 = opposite meaning (vectors point in opposite directions)
- We use cosine similarity because it focuses on the direction of vectors rather than their magnitude, making it ideal for comparing semantic meaning. Even if one sentence is longer than another, cosine similarity captures whether they express similar concepts.

Confidence Threshold (0.75)

- The threshold of 0.75 acts as a decision boundary:
 - If similarity $\geq 0.75 \rightarrow$ Accept as valid command
 - If similarity $< 0.75 \rightarrow$ Reject as unrecognized
- This threshold was chosen to balance two competing goals:
 - **Higher threshold** : Fewer false positives but misses valid variations
 - **Lower threshold** : Catches more variations but accepts wrong commands
- A threshold of 0.75 provides optimal balance, accepting semantic variations while rejecting unrelated phrases.
- Example:
 - "ışığı aç" vs "lambayı aç" $\rightarrow$ similarity: 0.87 (ACCEPTED)
 - "ışığı aç" vs "müzik çal" $\rightarrow$ similarity: 0.42 (REJECTED)

N-gram Phrase Detection

The n-gram model uses Gensim's Phrases and Phraser to detect and combine frequently co-occurring words into single tokens (e.g., "ışığı" + "aç" $\rightarrow$ "ışığı_aç"). This treats complete commands as atomic semantic units, preventing false matches based on shared verbs alone.

Limitations:

- Out-of-vocabulary words cause zero vectors, leading to false negatives
- Semantic similarity causes false positives (e.g., "pencereyi aç" matches "perdeyi aç")
- N-grams require exact phrase structure, reducing flexibility
- Performance depends on pre-trained Turkish model quality
- Threshold tuning needed for different domains
- Turkish-specific challenges (agglutination, vowel harmony)

CONCLUSION

- **Word-level embeddings** (Methods 1 & 3) achieved the best overall performance with F1-Score of 0.780, demonstrating strong recall (84.2%) while maintaining acceptable precision (72.7%).
- **TF-IDF weighting** provided no measurable improvement over simple mean vectors in our domain, likely because smart home commands are short and have similar term distributions.
- **N-gram phrase detection** (Methods 2 & 4) improved precision to 75.0% by treating commands as atomic units, but significantly reduced recall to 63.2% due to strict matching requirements.
- The **trade-off between precision and recall** is clear: word-level models accept more user variations (better user experience) while n-gram models are more conservative (fewer false activations).

REFERENCES

- Turkish Word2Vec: <https://github.com/akoksal/Turkish-Word2Vec>
- Gensim Documentation: <https://radimrehurek.com/gensim/>
- Mikolov et al. (2013). "Efficient Estimation of Word Representations"